

Total No. of Questions :6]

SEAT No. :

P108

[Total No. of Pages :2

APR - 18/ BE/ Insem. - 76

B.E. (Information Technology)

DISTRIBUTED SYSTEM

(2012 Course) (Semester - II) (414461)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1, or Q.2, Q.3, or Q.4, Q.5 or Q.6.
- 2) Figures to right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Explain the models based on fundamental properties that are common in all of the architectural models. [6]

b) What is Scalability? Explain the design challenges of Scalable Distributed Systems. [4]

OR

Q2) a) Explain distributed multimedia systems in brief and justify with example that webcasting is an examples of distributed multimedia systems. [6]

b) Define Transparency and Briefly Explain the various forms of Transparency in a Distributed Systems. [4]

Q3) a) Define the concept of Socket. Explain the issues and failure model with respect to Stream Communication. [6]

b) Explain the Message Queues considering programming model offered by message queueing systems. [4]

OR

Q4) a) Define the concept of Socket. Explain the issues and failure model with respect to Datagram Communication. [6]

b) What is Marshalling? How marshalling and serialization is used in communication between a client and a server? [4]

P.T.O.

Q5) a) Explain the following with respect to Distributed Object Middleware framework: [6]

- i) Inter-object communication
- ii) Lifecycle management
- iii) Activation and Deactivation.

b) State the applications of Web Services. Differentiate between SOAP and REST Web Services. [4]

OR

Q6) a) What are the issues with object-oriented middleware? [6]

b) Explain two main characteristics of distributed event-based systems. [4]